SOFTWARE REQUIREMENTS SPECIFICATION

Academic Courses & Schedule Planner

Course: CSAI 203 — Introduction to Software Engineering

Instructor: Dr. Mohamed Sami Rakha

Semester: Fall 2026

Phase: Phase 2 — Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the intended behavior, scope, users, functional capabilities, quality requirements, external interfaces, and conceptual domain model of the Academic Courses & Schedule Planner. It establishes a shared reference for the student team, teaching assistants, and evaluator during the design, implementation, and testing phases.

1.2 Scope

The Academic Courses & Schedule Planner is a web application intended to help students plan an academic semester. A student can browse course offerings, review course details and prerequisites, select available sections, build a timetable, detect overlapping meeting times, view academic records, and calculate a GPA from recorded grades. An administrator manages course information, semester offerings, section capacities, meeting times, prerequisite relationships, and published availability.

The system aims to reduce manual planning effort and help students identify invalid course selections before finalizing a personal plan.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification.
GPA	Grade Point Average, calculated from course credit hours and grade-point values.
Prerequisite	A course or condition that must be satisfied before a student may take another course.
Section	A specific offering of a course in a semester, with capacity and meeting times.
Timetable	A weekly view of the selected sections and their meeting times.
Functional Requirement (FR)	A behavior or service the system must provide.
Non-Functional Requirement (NFR)	A measurable quality or constraint imposed on the system.

MVC	Model–View–Controller, the required architectural pattern for this project.
Conflict	An overlap between two scheduled course meetings.

1.4 References

1) CSAI 203 Course Project Requirements, Fall 2026, Dr. Mohamed Sami Rakha
2) Lecture slides containing the "UML Use Case IEEE Description Template" required for final use-case descriptions; the exact lecture template must be checked before submission.

2. Overall Description

2.1 Product Perspective

The product is a standalone web-based academic planning application for a course-project environment. It is not assumed to be connected to a live Student Information System (SIS), official registration portal, or external identity provider. Course and academic data are maintained within the application by authorized users or seeded test data.

2.2 Product Functions

1) Create an account and authenticate users.

2) maintain a student academic profile and selected academic program.

3) Browse, search, and filter the course catalog and semester offerings.

4) Display course descriptions, credit hours, prerequisites, sections, meeting times, and availability.

5) Validate prerequisite completion before adding a course to a plan.

6) Add and remove course sections from a personal semester plan.

7) Detect overlapping meeting times and explain the conflicting entries.

8) Display a weekly timetable and allow the student to revise the plan.

9) Maintain completed courses and grades for planning purposes.

10) Calculate GPA using the configured grading scale and course credit hours.

11) Allow administrators to manage courses, prerequisites, sections, capacities, and semester publication.

2.3 User Classes and Characteristics

User class	Main goals	Expected characteristics
Student	Plan a semester, check prerequisites and conflicts, review records, calculate GPA.	Basic web-app skills; may be unfamiliar with academic planning rules.
Administrator	Maintain accurate catalog, offerings, prerequisites, capacities, and meeting times.	Authorized user familiar with course data and administration tasks.
Teaching Assistant	Review the deliverable and demonstrate that requirements are testable.	Uses the application and project documentation for evaluation

2.4 Software Stakeholders

Stakeholder	Interest in the system
Students	A clear and reliable way to build a feasible timetable and understand course eligibility.
Course project team	A scope that can be designed, implemented, tested, documented, and explained.
Instructor and TAs	Requirements that are detailed, traceable, testable, and consistent with course rules.
Academic department (assumed stakeholder)	Accurate course, prerequisite, capacity, and timetable information.
System administrator	Controlled access to administrative functions and data integrity.

2.5 Operating Environment

The application is intended to run in a modern desktop or mobile web browser with network access to the application server. The system utilizes a relational database and is deployed on a standard web hosting platform. It is fully accessible via modern desktop and mobile browser

2.6 Design and Implementation Constraints

1)The required SRS outline and scope in the CSAI 203 project requirements must be followed.

2)the project architecture must use the Model–View–Controller (MVC) pattern; its application will be described further in the Design Document.

2.7 User Documentation

The project should provide a short user guide or FAQ describing account access, searching the course catalog, adding and removing sections, understanding timetable conflicts, interpreting prerequisite warnings, and calculating GPA. A separate short administrator guide should explain course and section maintenance, capacity updates, prerequisite setup, and publishing offerings.

2.8 Assumptions and Dependencies

1)Users have access to a supported browser and a network connection to the application.

2)course offerings, meeting times, prerequisites, capacities, and grade-point mappings are entered accurately by authorized users or provided as approved test data.

3)The grading scale used for GPA calculations is explicitly configured; no particular university scale is assumed by this SRS.

4)Time slots are represented consistently, and meeting-time overlap is evaluated using start and end times.

5) The system does not automatically enroll does not automatically enroll students in an official university registration system.

6)Email verification, password recovery through email, live SIS integration, and third-party sign-on are outside the baseline scope

7)Users are expected to protect their credentials and sign out on shared devices.

3. Specific Requirements

3.1 Functional Requirements

FR(!) Account Registration

The system allow a new user to create an account by providing the required identity and login information. Before creating the account, it validate required fields, email format, password policy, and whether the email is already registered. If validation fails, the system identify the relevant field and shall not create a partial account. A successful registration create a user record with the default Student role

FR(2) Sign In and Sign Out

The system authenticate a registered user using valid credentials and create an authenticated session. Invalid credentials produce a general error that does not reveal whether a specific account exists. Signing out invalidate the active session so that protected pages cannot be accessed using the signed-out session.

FR(3) Role-Based Access

The system distinguish Student and Administrator permissions. Students may manage their own academic plan and view their own records. Administrators may manage catalog data and semester offerings. If a user without permission requests a protected operation, the system reject the operation and not change stored data.

FR(4) Manage Student Profile

The system allow a student to view and update permitted profile fields such as name, student number, academic program, and current academic level. Required fields be validated. Changes shall be associated with the authenticated student and not allow the student to edit another student's profile.

FR(5) Browse Course Catalog

The system display available course records with course code, title, description, credit hours, and prerequisite information when available. If no courses match the current filters, the system display an empty-state message rather than an error.

FR(6) Search and Filter Courses

The system allow a student to search by course code or title and filter by relevant catalog attributes such as credit hours or semester availability. Search ignore leading and trailing whitespace and return only records matching the supplied criteria. Clearing filters restore the unfiltered catalog view.

FR(7) View Course and Section Details

The system show the selected course's description, credit hours, prerequisites, offered sections, meeting times, room or location when provided, and current capacity status. Missing optional data be clearly marked as unavailable rather than fabricated.

FR(8) Manage Semester Plan

The system allow a student to create or update a personal plan for a selected semester. A plan contain the selected course sections and be associated with its owner and semester. The system prevent duplicate addition of the same section to the same plan and confirm successful changes.

FR(9) Validate Prerequisites

Before a course is added to the plan, the system compare the course's prerequisite rules with the student's recorded completed courses and grades. If requirements are not satisfied, the system explain which prerequisite is missing and prevent the addition under the baseline policy. If prerequisite data is not configured, the system clearly indicate that eligibility could not be fully verified.

FR(10) Detect Timetable Conflicts

When a section is added or a plan is checked, the system compare the meeting times of selected sections. Two meetings conflict when they occur on the same day and their time intervals overlap. The system identify both affected sections and the overlapping time range. A non-conflicting selection not be reported as a conflict.

FR(11) Check Section Capacity

The system show whether a section is available based on its configured capacity and current enrollment count, if that count is maintained. A full section shall be flagged and not be added to the plan The system re-check capacity when a section is selected to reduce the chance of using stale availability data.

FR(12) Remove a Section from Plan

The system allow a student to remove a section from their own semester plan. It recalculate the plan's credit-hour total and refresh the timetable and conflict results. Removing a section not delete the underlying course or section record.

FR(13) Generate and Display Timetable

The system display selected section meetings in a weekly timetable organized by day and time. Each entry show course code or title, meeting time, and room when available. The timetable update after a successful plan change and visually distinguish detected conflicts.

FR(14)View Academic Records

The system allow a student to view completed courses and their recorded grades for planning purposes. Records be ordered or filterable by academic term. The student not be able to alter official-grade fields through the student interface

FR(15)Calculate GPA

The system calculate GPA using the configured grade-point mapping and the credit hours of eligible graded courses. The calculation use the weighted formula: sum of (course credit hours × grade points) divided by sum of credit hours included in the calculation. The system state which courses are included and handle the case of zero eligible credit hours without dividing by zero.

FR(16) Track Credit Hours and Progress

The system calculate and display total planned credit hours for the selected semester and, where the necessary academic data is available, completed credit hours. If the program's total required credits are configured, it may show a progress indicator based on completed credits; it label the result as an estimate when transfer credits or special rules are not represented.

FR(17) Manage Course Records (Admin)

The system allow an authorized administrator to create, view, update, and deactivate course records. It validate required course code, title, and credit-hour fields, and it prevent duplicate course codes. Deactivating a course preserve historical references and prevent it from being offered in new plans unless the administrator reactivates it.

FR(18) Manage Sections and Offerings (Admin)

The system allow an administrator to create and update semester offerings and sections, including semester, section identifier, meeting schedule, capacity, and room when available. It validate that capacity is not negative and that meeting start time precedes end time. Invalid data not be saved.

FR(19) Manage Prerequisite Rules (Admin)

The system allow an administrator to define and update prerequisite relationships between courses and, where needed, a minimum required grade. The system prevent a course from being configured as its own prerequisite and warn about circular prerequisite chains that could make course planning impossible.

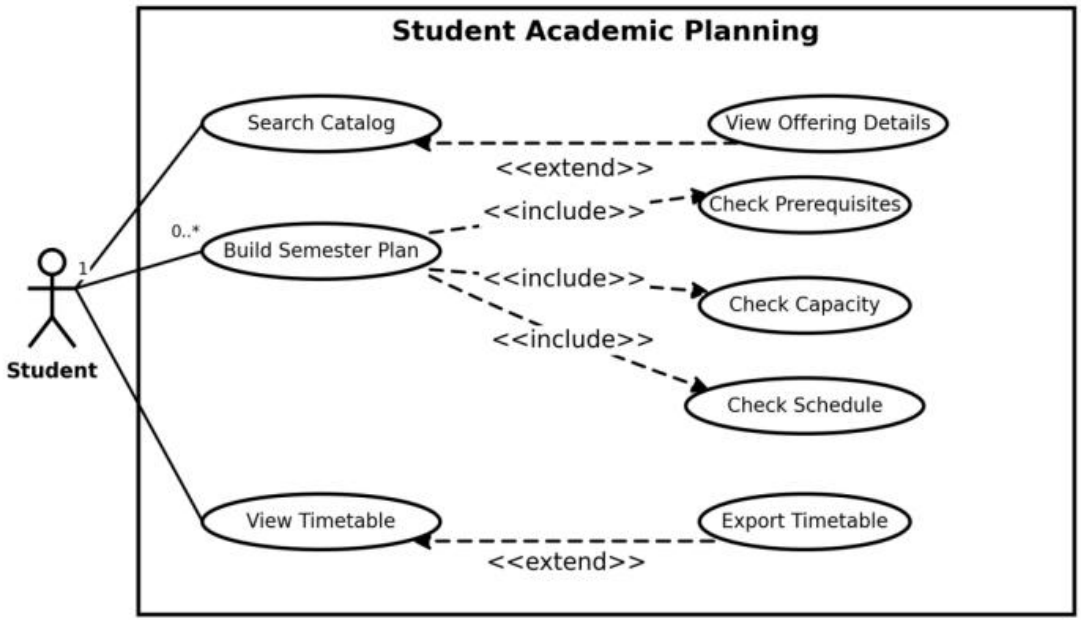
FR(20) Publish Offerings and Review Validation

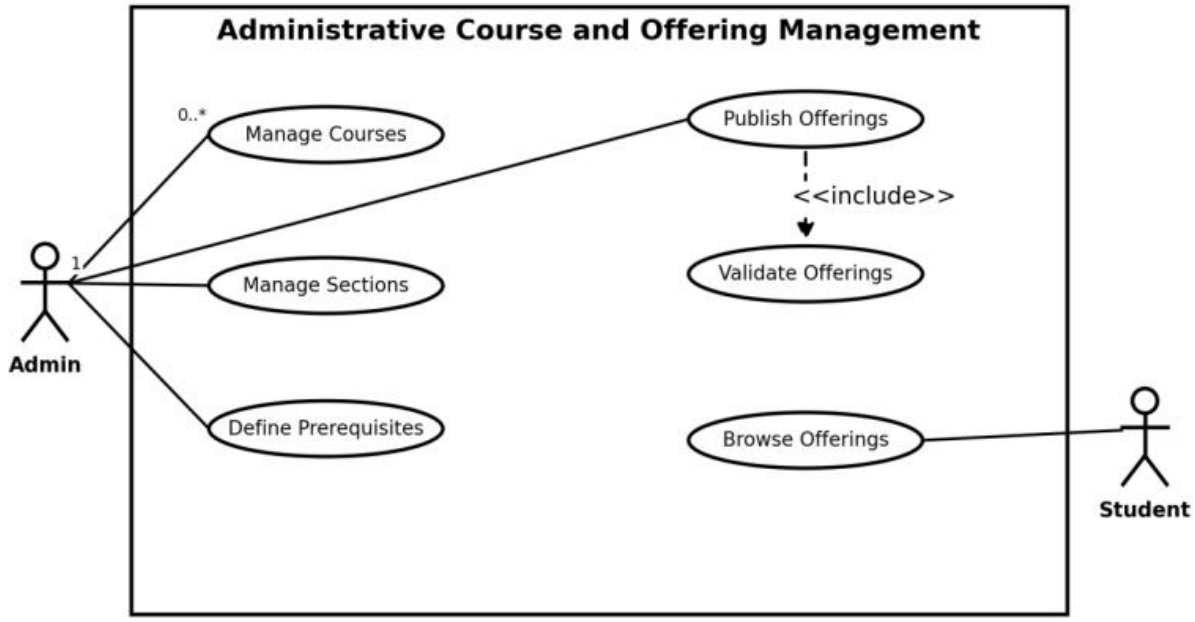
The system allow an administrator to publish semester offerings after required fields have been validated. Before publication, the system report incomplete course/section data, invalid meeting times, and suspicious prerequisite relationships. Unpublished change not appear as active offerings to students until publication is confirmed.

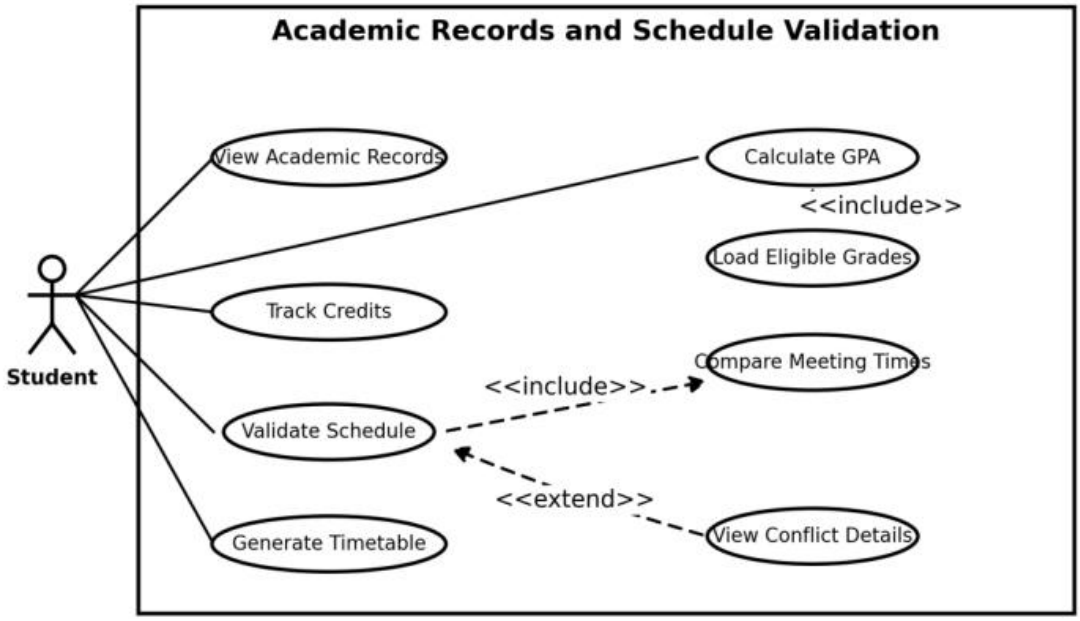
3.2 Use Case Model

The following figures divide the system's use cases into student planning, administrative course management, and academic records/schedule validation.

3.2.1 Use Case Diagrams







3.2.2 Use Case Description

UC-01 — Create Account

Field	Description
Primary actor	Student
Goal	A person wants to create a student account.
Preconditions	User is not signed in.
Main success scenario	1. User opens registration. 2. User enters required details. 3. System validates fields and checks for an existing email. 4. System creates the account with Student role. 5. System confirms registration.
Alternate / exception flows	Required fields invalid or email already exists: system displays a validation message and creates no account.
Postconditions	A student account is stored and the user can sign in.
Traceability	FR-01

UC-02 — Sign In

Field	Description
Primary actor	Student, Administrator
Goal	A registered user wants to access permitted functions.
Preconditions	The user has an existing account.
Main success scenario	1. User enters credentials. 2. System validates credentials. 3. System creates an authenticated session. 4. System opens the appropriate landing page for the user role.
Alternate / exception flows	Invalid credentials: system displays a general authentication error. Disabled account: access is denied.

Postconditions	Authenticated session is established only for a valid account.
Traceability	FR-02, FR-03

UC-03 — Build Semester Plan

Field	Description
Primary actor	Student
Goal	A student wants to select sections for a semester.
Preconditions	Student is authenticated; semester offerings are published.
Main success scenario	1. Student selects a semester. 2. System displays offerings. 3. Student selects a section. 4. System validates prerequisites and capacity. 5. System checks timetable conflicts. 6. System adds the section if all blocking rules pass. 7. System refreshes the plan summary.
Alternate / exception flows	Prerequisite failure, full section, duplicate section, or conflict: system explains the issue and applies the defined planning policy.
Postconditions	Plan contains the accepted section and the updated credit-hour total.
Traceability	FR-05–FR-13

UC-04 — Review Timetable Conflicts

Field	Description
Primary actor	Student
Goal	A student wants to understand schedule overlaps.

Preconditions	Student has a semester plan with at least one selected section.
Main success scenario	1. Student opens the timetable or requests validation. 2. System compares meetings on matching days. 3. System identifies overlapping intervals. 4. System displays affected sections and overlap times.
Alternate / exception flows	No selected sections: system displays an empty timetable. Missing meeting data: system flags that the schedule cannot be fully validated.
Postconditions	Student receives a timetable and an explicit conflict result.
Traceability	FR-10, FR-13

UC-05 — Calculate GPA

Field	Description
Primary actor	Student
Goal	A student wants a GPA estimate from recorded grades.
Preconditions	Student has academic records; grade-point mapping is configured.
Main success scenario	1. Student opens GPA calculation. 2. System selects eligible graded courses. 3. System multiplies each course's credit hours by its grade points. 4. System divides the weighted sum by eligible credit hours. 5. System displays GPA and included courses.
Alternate / exception flows	No eligible graded credits: system explains that GPA cannot be calculated yet and avoids division by zero.

Postconditions	GPA estimate and calculation basis are displayed.
Traceability	FR-14, FR-15

UC-06 — Maintain Course Offering

Field	Description
Primary actor	Administrator
Goal	An administrator wants to maintain course and section data.
Preconditions	Administrator is authenticated and authorized.
Main success scenario	1. Administrator opens course/offerings management. 2. Administrator creates or edits course/section fields. 3. System validates required fields, identifiers, times, capacity, and prerequisites. 4. Administrator saves the changes. 5. System confirms the result or reports validation errors.
Alternate / exception flows	Duplicate code, invalid time interval, negative capacity, or invalid prerequisite: save is rejected and the relevant error is shown.
Postconditions	Valid catalog/offering data is stored; invalid data is not committed.
Traceability	FR-17–FR-19

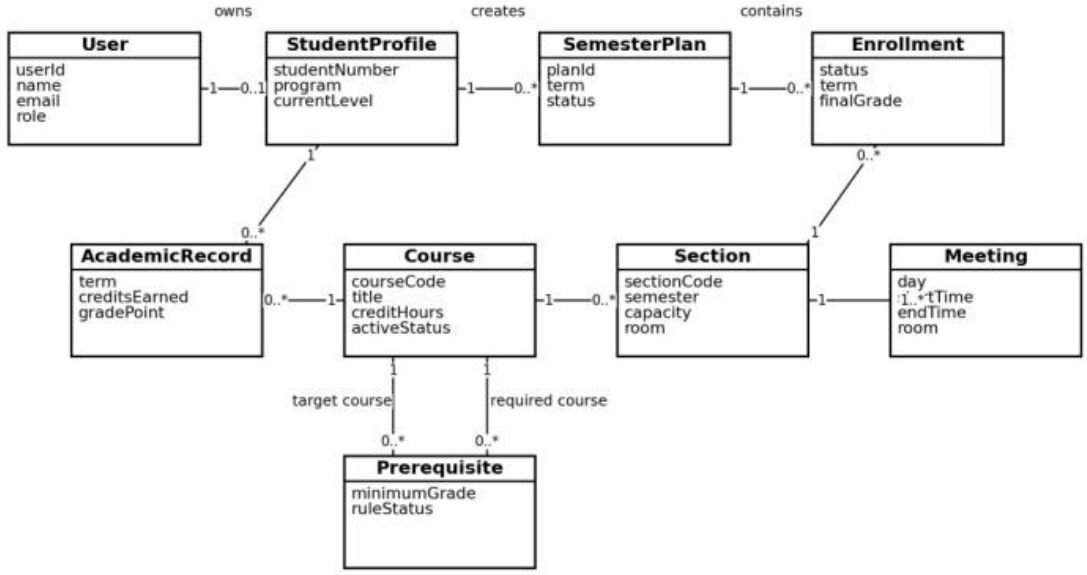
UC-07 — Publish Semester Offerings

Field	Description
Primary actor	Administrator
Goal	An administrator wants to make semester sections visible to students.

Preconditions	Administrator is authenticated; offerings have been entered.
Main success scenario	1. Administrator requests validation. 2. System checks required fields and prerequisite relationships. 3. System presents validation findings. 4. Administrator resolves blocking issues. 5. Administrator confirms publication. 6. System marks the offerings as published.
Alternate / exception flows	Blocking validation issues: publication is prevented and the administrator receives a report.
Postconditions	Validated offerings become visible in the student catalog.
Traceability	FR-20

3.3 Domain Model

3.3.1 Conceptual Class Diagram



3.3.2 Class Descriptions

Class	Purpose, attributes, and relationships
User	Person who signs in. Attributes: userId, name, email, passwordHash, role, accountStatus. Relationship: may own 0..1 StudentProfile; role controls access.
StudentProfile	Student-specific academic context. Attributes: studentNumber, program, currentLevel. Belongs to one User and is associated with plans, enrollments, and academic records.
SemesterPlan	The student's plan for one term. Attributes: planId, term, status. Belongs to a StudentProfile and contains plan selections.
Enrollment	Planned or completed course/section participation. Attributes: status, term, finalGrade. Connects a plan/section with a student and records selection or completion status
AcademicRecord	Completed academic result. Attributes: term, creditsEarned, gradePoint. Belongs to one StudentProfile and supplies eligible data for GPA/progress calculations.
Course	Catalog course. Attributes: courseCode, title, description, creditHours, activeStatus. May be offered through multiple Sections and may participate in prerequisite rules.
Section	A specific course offering in a term. Attributes: sectionCode, semester, capacity, room, publicationStatus. Belongs to one Course and has Meeting records.
Meeting	A scheduled session for a Section. Attributes: day, startTime, endTime, room. Belongs to

	one Section; times are compared for conflict detection.
Prerequisite	A rule linking a target Course to a required Course. Attributes: minimumGrade, ruleStatus. The relationship must not create a course requiring itself.

3.4 Non-Functional Requirements

NFR-01 Response Time: For 500 courses, 2,000 sections, and 100 concurrent users, ≥95% of catalog/plan/timetable requests shall finish within 2 seconds. Test with a repeatable load test; pass if the threshold is met and no unhandled errors occur.

NFR-02 Security and Access Control: Protected actions require authentication and role/ownership checks; passwords must be hashed. Test anonymous, cross-student, and student-to-admin access plus stored credentials; pass if unauthorized actions are denied without data changes and no plain-text password exists.

NFR-03 Data Integrity: Reject invalid course codes, negative capacity, invalid meeting ranges, duplicate section additions, and self-prerequisites. Test invalid inputs and saved records; pass if all are rejected without partial or inconsistent data.

NFR-04 Usability and Learnability: A first-time student completes search, add-section, timetable, and conflict-identification tasks unaided. Test 5 users; pass if at least 4 complete all tasks and correctly explain a conflict.

NFR-05 Compatibility and Responsive Layout: Core workflows work in current Chrome and Edge on desktop and at 375px mobile width. Test the same workflow on all targets; pass if no blocking functional or layout defects occur.

NFR-06 Availability and Error Handling: During a 30-minute demo, core workflows remain usable and recoverable failures show controlled messages. Test normal tasks plus invalid forms/simulated failures; pass if no induced failure creates an unhandled error page.

3.5 External Interface Requirements

3.5.1 User Interface

1)Registration and sign-in screens shall provide clearly labeled fields, validation feedback, and a visible route to the relevant account action.

2)The student dashboard shall summarize the selected semester plan, total planned credit hours, and any detected blocking issues.

3)The catalog view shall support search/filtering and provide a course detail view containing prerequisites and section information.

4)The semester planner shall show selected sections and allow adding/removing sections with confirmation or clear status messages.

5)The timetable view shall organize meetings by day and time, identify overlapping meetings, and remain usable at mobile widths.

6)The GPA view shall display the calculated result, the included courses/credits, and a note about the configured grade-point mapping.

7)The administrator interface shall provide forms for courses, sections, capacities, prerequisites, and publication, with validation results.

3.5.2 Hardware Interface: No specialized hardware interface is required. The system uses standard computers, tablets, or smartphones through a web browser. Printing a timetable is optional and may use the browser's standard print function; no dedicated printer integration is required.

3.5.3 Software Interface: Store users, courses, sections, meetings, prerequisites, semester plans, and academic records in a persistent database selected by the team. External university SIS, identity-provider, and GPA-service integrations are outside the baseline version.

3.5.4 Communication Interface: The browser communicates with the application server using HTTP during local development and HTTPS for any deployment over an untrusted network. The system should use secure session handling and avoid exposing credentials or sensitive academic data in URLs or diagnostic messages. If client/server requests are implemented through APIs, request and response formats and validation errors should be consistent and documented.

4. Appendices

4.1 Appendix A: Data Dictionary

Proposed data entity / table	Purpose and key data
Users	Login identity, password hash, role, and account status.
StudentProfiles	Student number, program, level, and linked user.

Courses	Course code, title, description, credit hours, and active status.
Sections	Course, section code, term, capacity, room, and publication status.
Meetings	Section, day, start/end time, and room.
CoursePrerequisites	Target course, required course, and minimum grade.
SemesterPlans	Student, term, and plan status.
Enrollments	Plan/student linked to a section; status and grade if applicable.
AcademicRecords	Term-level GPA/credits summary, if stored.
GradeScale	Mapping from grades to numeric GPA points.

4.2 Appendix B: Glossary

Term	Meaning in this SRS
Academic term	Semester or defined study period.
Capacity	Maximum seats allowed in a section.
Conflict-free timetable	No overlapping meeting intervals on the same day.
Grade-point mapping	Converts a grade to numeric points for GPA.
MVC	Model-View-Controller, the required architecture.
Published offering	Validated section visible to students.

Traceability	Links requirements to design, code, and tests.
Use case	Actor interaction to achieve a goal.

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